American Computer Science League

2024-2025 ● Contest 3: Short Problems ● Intermediate Division

1. Boolean Algebra

How many ordered triples make this expression TRUE?

$$A(\overline{B} + C) + \overline{AB} + C$$

A. 1
B. 3
C. 4
D. 5
E. 7

2. Boolean Algebra

Simplify this expression:

$$\overline{A}B(\overline{C + A}) + A\,B(\overline{\overline{A} + C})$$

A. $B + \overline{C}$
B. $\overline{A}B\overline{C}$
C. $A + \overline{B}C$
D. $B\overline{C}$
E. $A\overline{B}C$

3. Data Structures

List the nodes at depth 2 from left to right in the binary search tree for:

RUGBYSEVENS

A. B, E, N, S, Y
B. B, N, S, Y
C. E, N, S, Y
D. B, N, S, V
E. E, N, S

4. Data Structures

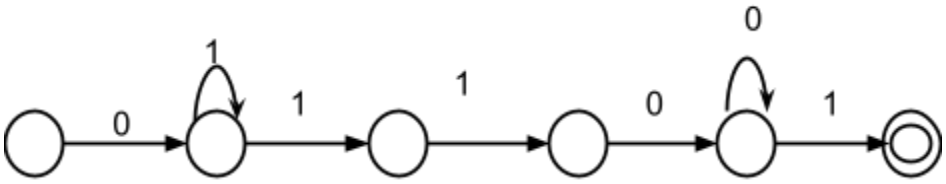
Given an initially empty stack and the following commands on the stack, what item will be popped next?

```
PUSH("A")
PUSH("R")
PUSH("C")
x = POP()
PUSH("H")
x = POP()
x = POP()
PUSH("E")
PUSH("R")
PUSH("Y")
x = POP()
x = POP()
```

A. A
B. C
C. E
D. R
E. H

5. FSAs

How many of the following strings satisfy the FSA below?



a. 01111101 b. 011101 c. 0111111

d. 0111001 e. 0110 f. 011110001

A. 1
B. 2
C. 3
D. 4
E. 5

6. Regular Expressions

Which of the following strings match the regular expression below?

[J-PT-V]*[^c^f-m]*[2468]

a. Jjp2 b. KULaazzz24 c. L6

d. TTOOB8 e. dstxyz4 f. 4

A. a, c, f
B. b, d, e
C. c, d, e, f
D. a, b
E. b, e, f

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1. 布尔代数

请问有多少个有序三元组可以使得**下列**表达式的结果为真？

$$A(\overline{B} + C) + \overline{AB} + C$$

A. 1
B. 3
C. 4
D. 5
E. 7

2. 布尔代数

请**简化下列表达式：**

$$\overline{A}B(\overline{C + A}) + A\,B(\overline{\overline{A} + C})$$

A. $B + \overline{C}$
B. $\overline{A}B\overline{C}$
C. $A + \overline{B}C$
D. $B\overline{C}$
E. $A\overline{B}C$

3. 数据结构

请按从左到右的顺序列出**下列字符串构成的二叉搜索树中深度为 2 的节点：**

RUGBYSEVENS

A. B, E, N, S, Y
B. B, N, S, Y
C. E, N, S, Y
D. B, N, S, V
E. E, N, S

4. 数据结构

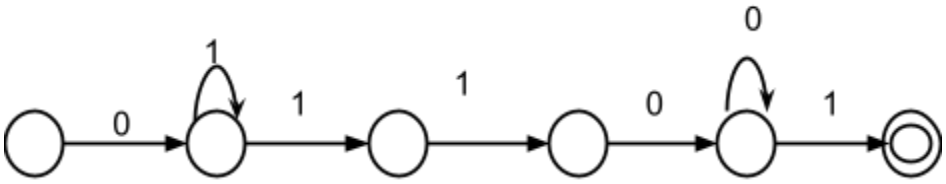
给定一个初始为空的栈，以及**下列**对**该**栈的操作指令，**请问**下一个被弹出的元素是什么？

```
PUSH("A")
PUSH("R")
PUSH("C")
x = POP()
PUSH("H")
x = POP()
x = POP()
PUSH("E")
PUSH("R")
PUSH("Y")
x = POP()
x = POP()
```

A. A
B. C
C. E
D. R
E. H

5. FSAs

请问有多少个字符串满足**下列** FSA？



a. 01111101　　b. 011101　　c. 0111111

d. 0111001　　e. 0110　　f. 011110001

A. 1
B. 2
C. 3
D. 4
E. 5

6. 正则表达式

请问哪些字符串与**下列**正则表达式匹配？

[J-PT-V]*[^c^f-m]*[2468]

a. Jjp2　　b. KULaazzz24　　c. L6

d. TTOOB8　　e. dstxyz4　　f. 4

A. a, c, f
B. b, d, e
C. c, d, e, f
D. a, b
E. b, e, f